# Patchify architecture — PatchEmbed3DPlus1D

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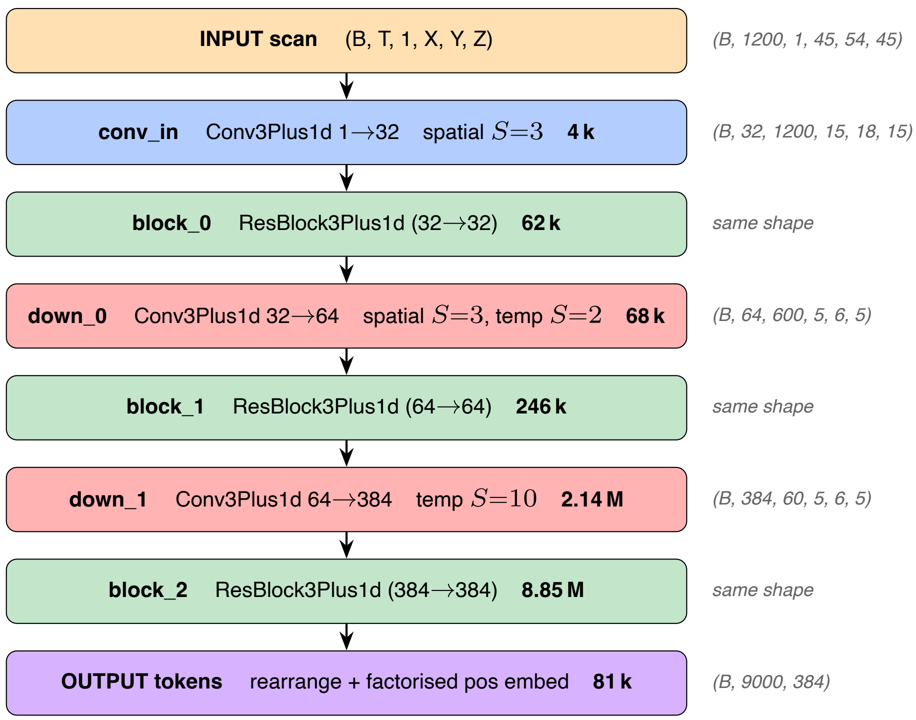
**Hierarchical 3D+1D patchify encoder, inspired by MovieGen TAE** (github.com/MathieuTuli/MovieGen, `tae.py:740`). Basic block = `Conv3Plus1d` = Conv3d spatial (per frame) + Conv1d temporal (per voxel-position) — separable substitute for the missing `Conv4d`.




**Total patchify parameters : 11.45 M**

**Setup HCP-YA** — $T{=}1200$, temporal_kernel $=20$, image $45 \times 54 \times 45$, patch_size $=9$.

**Strides totaux** : spatial $3 \cdot 3 \cdot 1 = 9$ (= patch_size), temporal $1 \cdot 2 \cdot 10 = 20$ (= temporal_kernel).

**Grille de tokens** : $T_{\text{eff}} = 60$, $N_{\text{spatial}} = 5 \cdot 6 \cdot 5 = 150$ □ **9 000 tokens** dim 384.

**Pos embed factorisé** : `pos_temporal(60×384)` + `pos_spatial(150×384)` + `pos_cls(1×384)` = 81 k params, vs 3.46 M pour une table plate (43× moins).